

# Haihao Lin

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## Research Interests

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VLA post-training stability; representation-preserving fine-tuning; embodied AI; representation learning; deployment-aware model adaptation.

## Summary

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First- and corresponding-author work on VLA post-training stability. I identify *action-fiber residual collapse* during VLA fine-tuning and propose **FiberTune**, a training-time, inference-free residual-preservation objective, validated across two backbones ( $\pi_{0.5}$ , OpenVLA-OFT), two simulation benchmarks (CALVIN, LIBERO), and a real SO-101 robot.

## Education

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**M.S., Computer Science (AI)**, University of Chinese Academy of Sciences (UCAS)   2024.09 – 2027.06 (expected)

Advisor: Prof. Xiangsheng Huang. Focus: embodied AI / VLA / representation learning.

**B.E., Computer Science & Technology**, Jiangsu University

2020.09 – 2024.06

GPA rank 1/171.

## Publications

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**FiberTune: Preserving Action-Fiber Visual Residuals in Vision-Language-Action Fine-Tuning.** CoRL 2026 (under review; **first & corresponding author**). [arXiv:2606.08653](https://arxiv.org/abs/2606.08653) · [fiber-tune.github.io](https://fiber-tune.github.io)

- Positive in all six controlled settings; CALVIN ABC→D success rate 61.4→72.1% (+10.7 pp); real SO-101 task success 72.7→78.1%.

## Research Experience

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- **VLA representation-preserving fine-tuning (FiberTune)** — problem formulation (action fibers), method design, cross-backbone/benchmark experiments, residual diagnostics.
- **Real-robot evaluation (SO-101 pick-place)** — real-robot data collection and OOD (held-out color) evaluation pipeline.

## Skills

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- **Languages:** Python, C/C++.
- **ML & VLA:** PyTorch; custom training-time objectives; fine-tuning pipelines for  $\pi_{0.5}$  (LeRobot) and OpenVLA-OFT.
- **Evaluation:** CALVIN chain-eval, LIBERO, real SO-101; multi-seed with confidence intervals.
- **Robotics & Systems:** real-robot deployment & data collection; multi-GPU training & experiment orchestration.

## Honors

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- Undergraduate GPA rank **1/171**; Outstanding Graduate; Jiangsu University President's Medal (2022–23); Provincial Merit Student (Jiangsu).
- National Scholarship ×2 (2021–22, 2022–23); First-class Scholarship; Merit Student (multiple).
- Outstanding undergraduate thesis.

## Competitive Programming

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- ICPC Asia Regional — Silver ×3 (Hangzhou 2023, Hangzhou 2024, Shenyang 2024); Asia East-Continent Final 2024 (1st in bronze tier).
- Lanqiao Cup National Finals — First Prize ×2 (C/C++ 2023; Python 2024, 9th nationally).